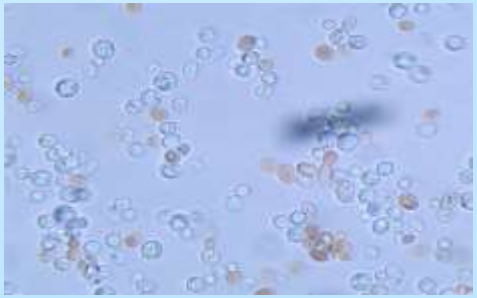
Kingdom Fungi

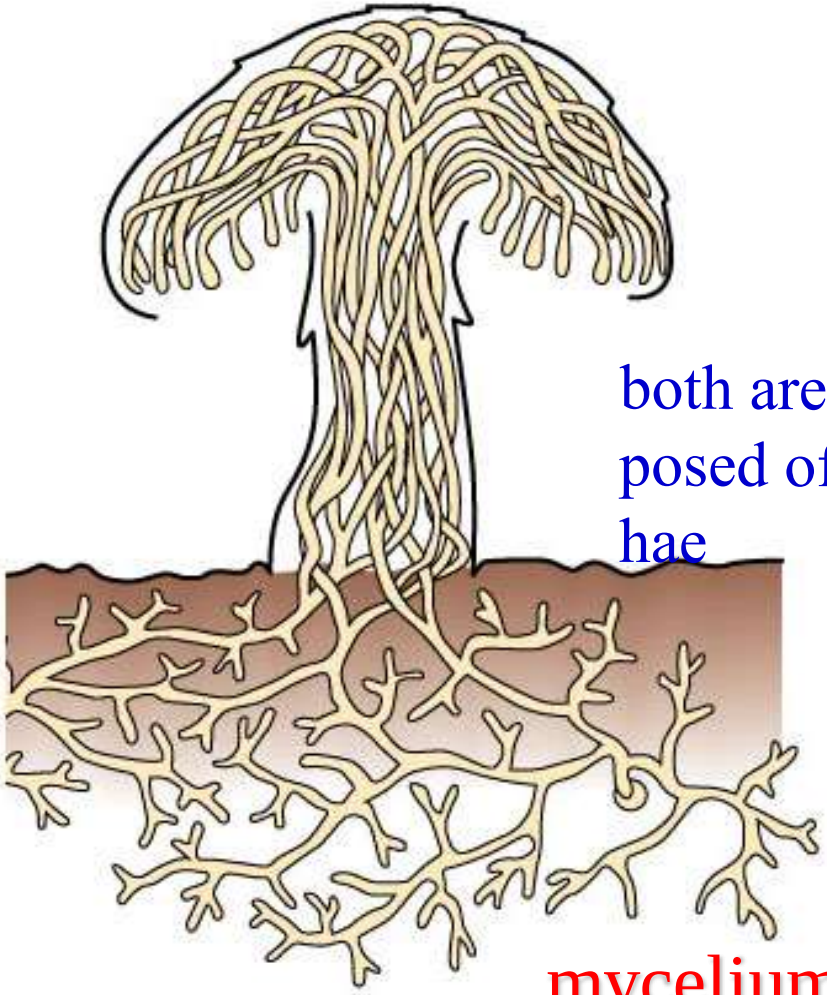
- The characteristics of fungi
- The evolution of th e fungi
- Fungal classificatio n
- Fungal life cycles

The Characteristics of Fungi

- Body form
  - * unicellular
  - * filamentous (tube-like strands called hypha (singular) or hyphae (plural)
  - * mycelium = aggregate of hyphae
  - * sclerotium = hardened mass of mycelium that generally serves as an overwintering stage.
  - * multicellular, such as mycelial cords, rhizomorphs, and fruit bodies (mushrooms)



fruiting bodies



both are com
posed of hyp
hae

mycelium

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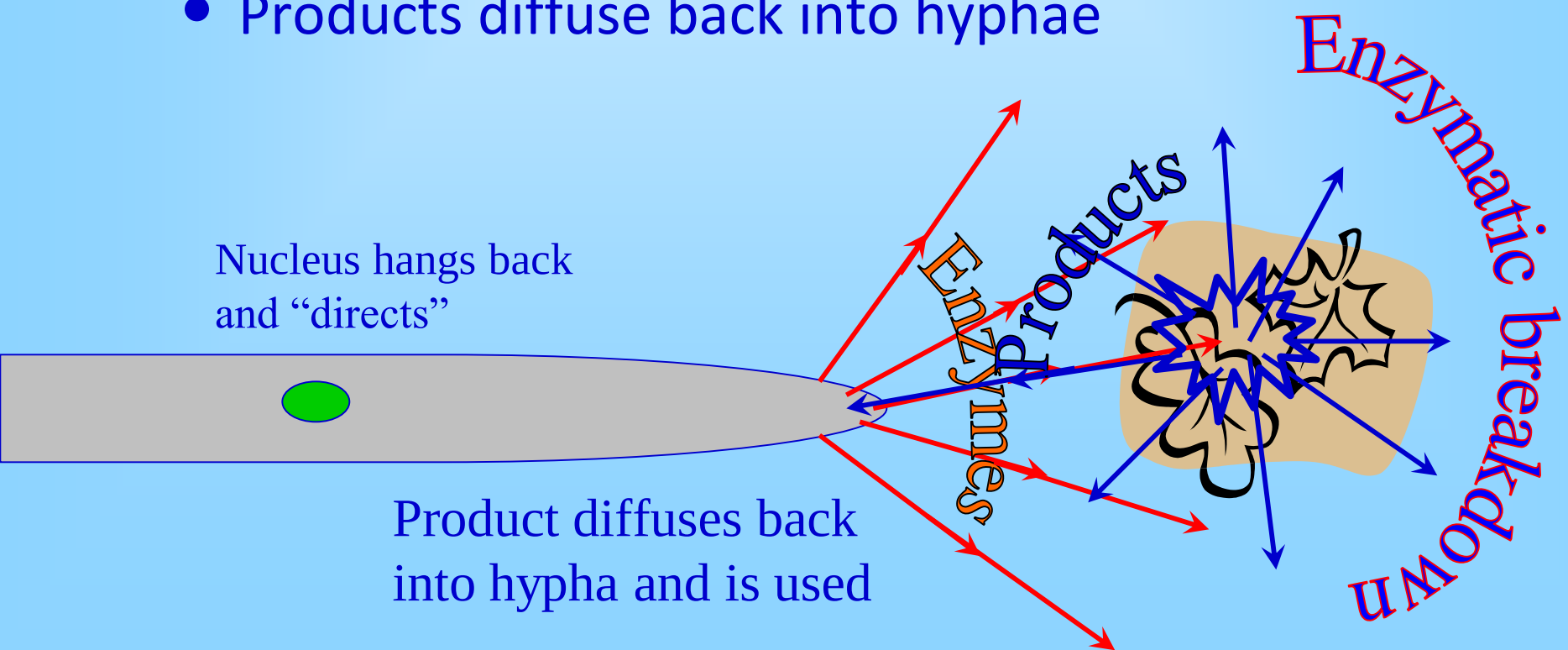
The Characteristics of Fungi

- Heterotrophy - 'other food'
  * Saprophytes or saprobes - feed on dead ti ssues or organic waste (decomposers)
  * Symbionts - mutually beneficial relations hip between a fungus and another organis m
  * Parasites - feeding on living tissue of a ho st.
    - Parasites that cause disease are called p athogens.

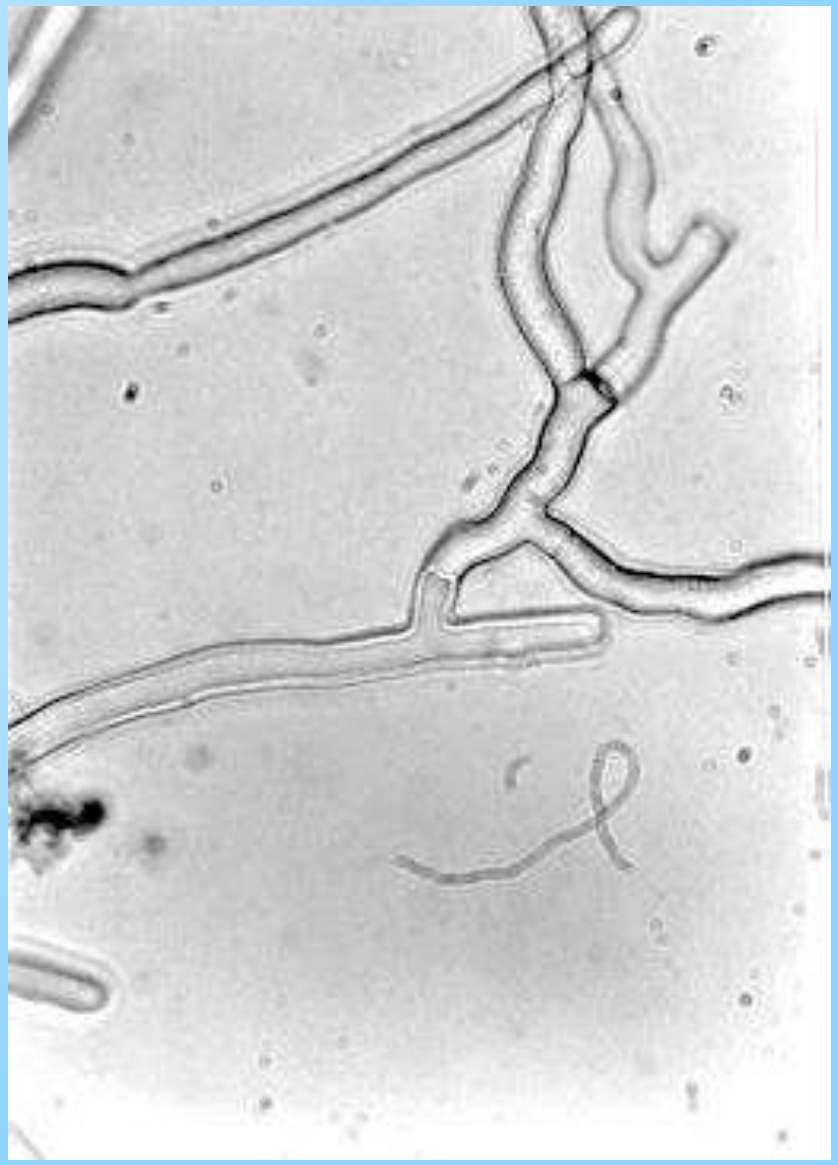
Heterotrophic by Absorption

- Fungi get carbon from organic sources
- Hyphal tips release enzymes
- Enzymatic breakdown of substrate
- Products diffuse back into hyphae



Hyphae

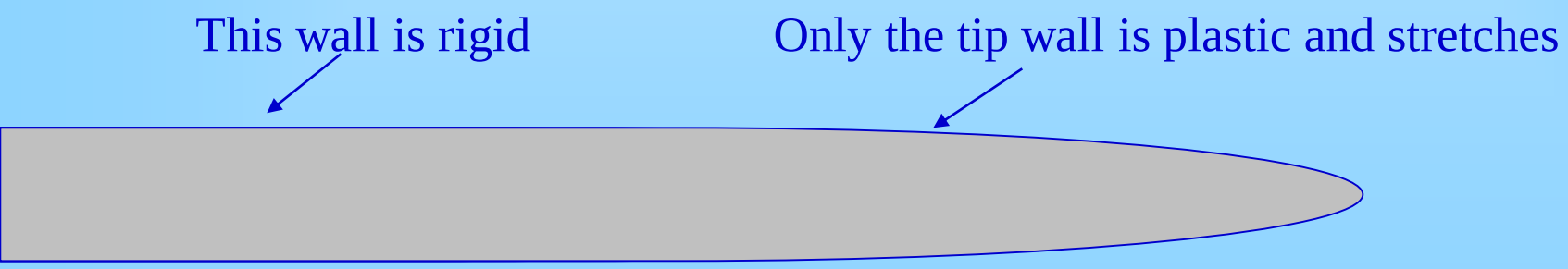
- Tubular
- Hard wall of chitin
- Crosswalls may form compartments (± cells)
- Multinucleate
- Grow at tips



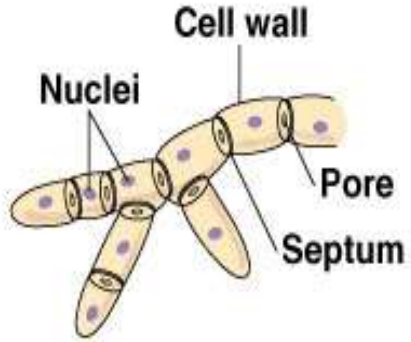
Hyphal growth

- Hyphae grow from their tips
- Mycelium = extensive, feeding web of hyphae

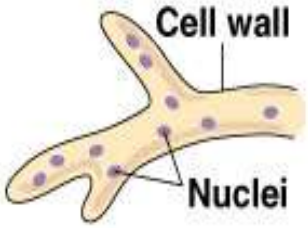
- Mycelia are the ecologically active bodies of fu ngi



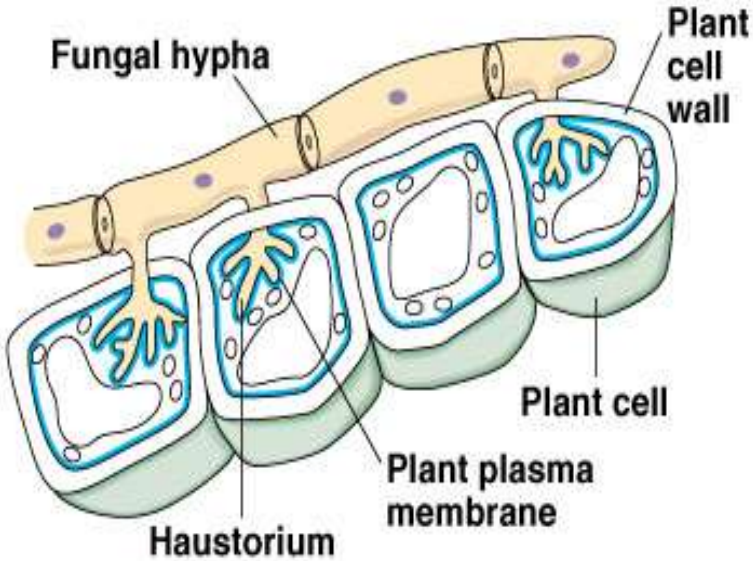
Modifications of hyphae



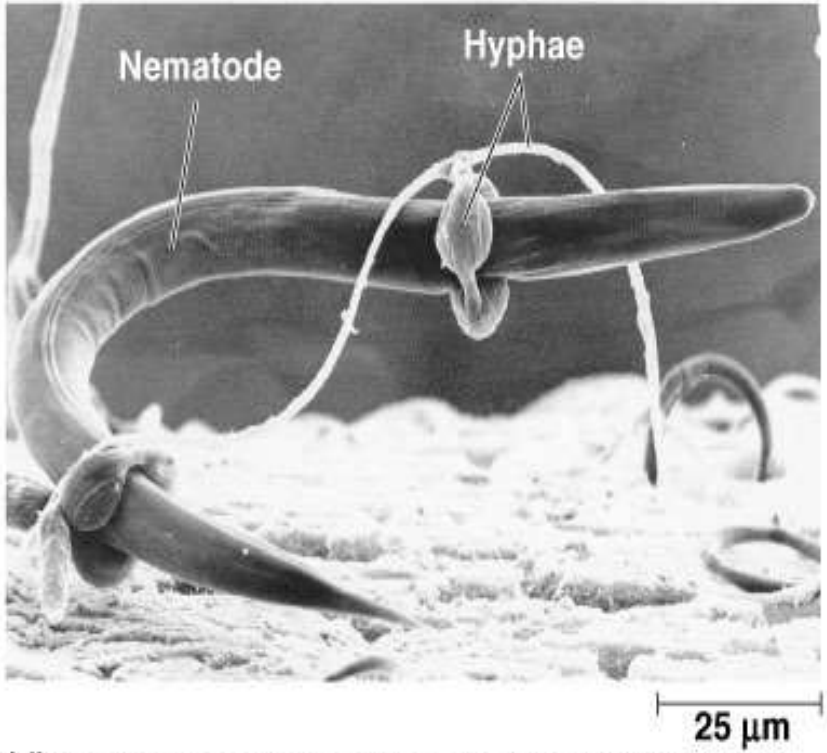
(a) Septate hypha



(b) Coenocytic hypha



(c) Haustoria



(d) Hyphae adapted for trapping and killing prey

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Fungi as Saprobes and Decom posers

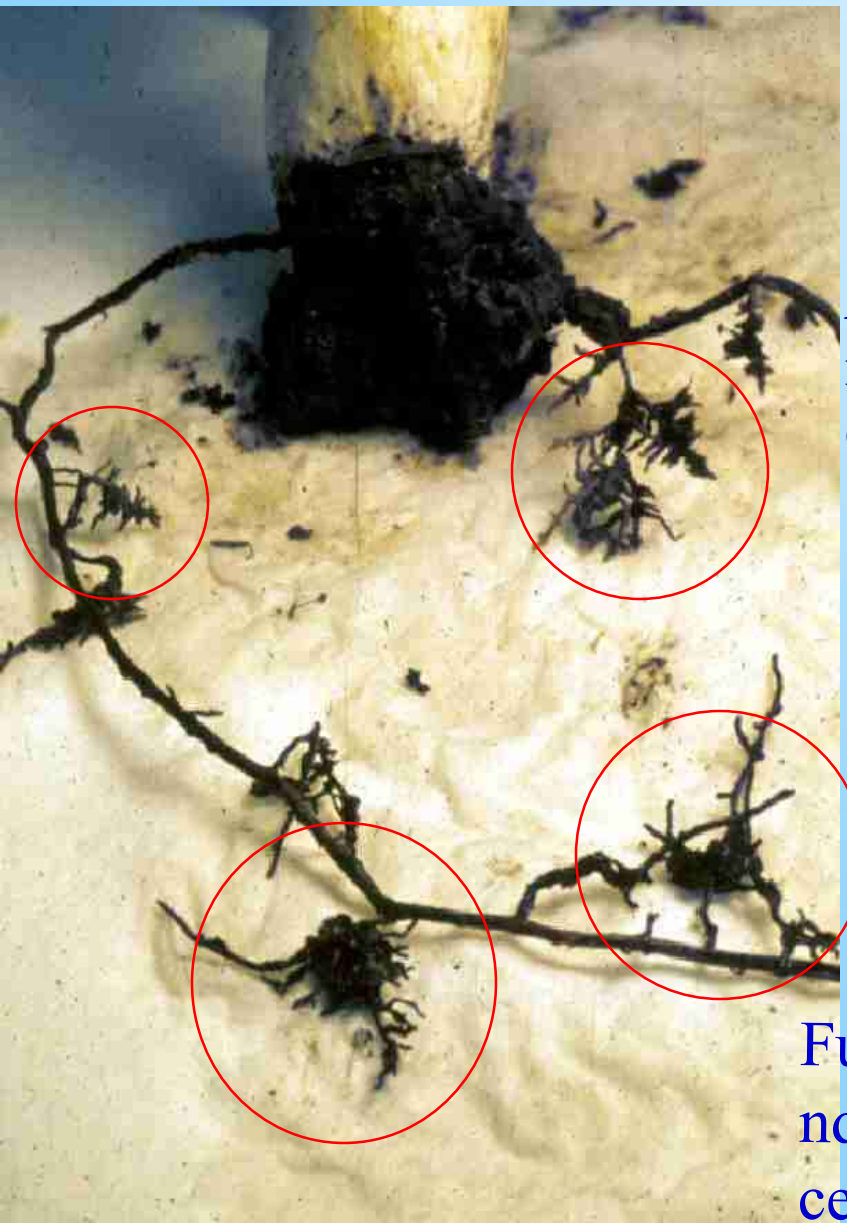
Fungi as Symbionts (Mutualism)

Mycorrhizae

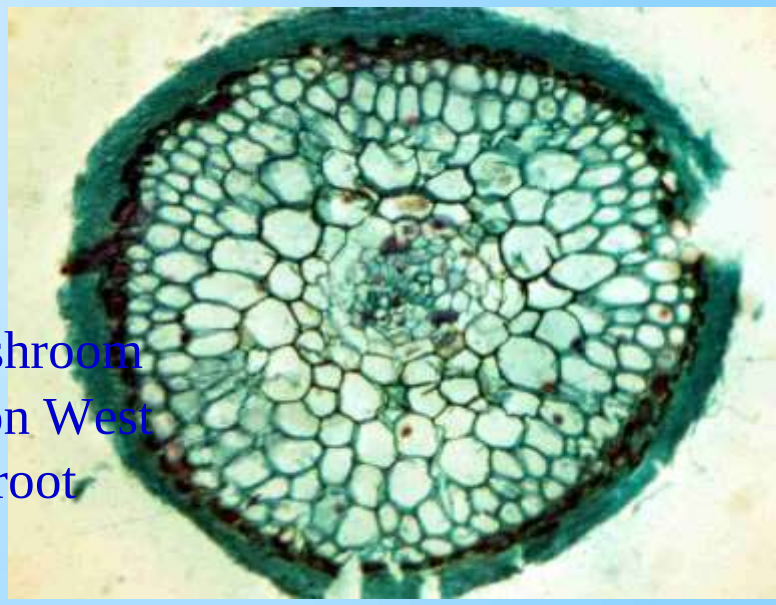
- "Fungus roots"
- Mutualism between:
  - Fungus (nutrient & water uptake for plant)
  - Plant (carbohydrate for fungus)
- Several kinds
  - Zygomycota – hyphae invade root cells
  - Ascomycota & Basidiomycota – hyphae invade root but don't penetrate cells

- **Extremely** important ecological role of fungi!

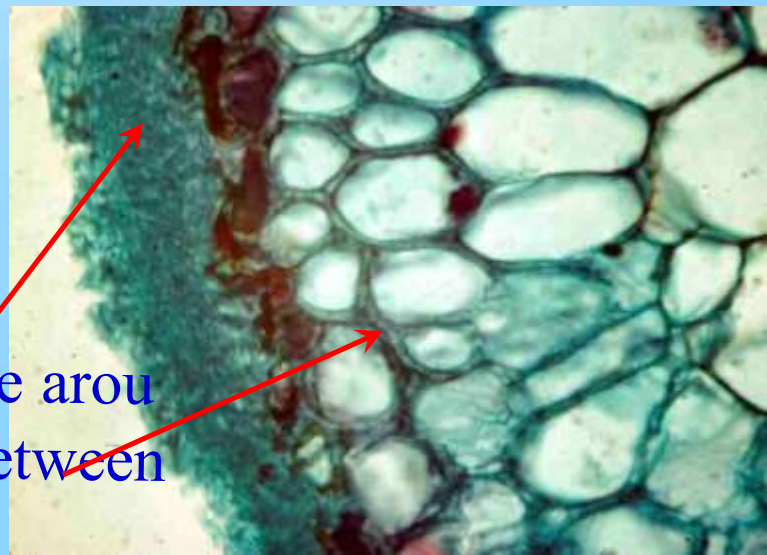
“Ecto”mycorrhizae



Russula mushroom mycorrhizas on West ern Hemlock root



Mycorrhiza cross sections

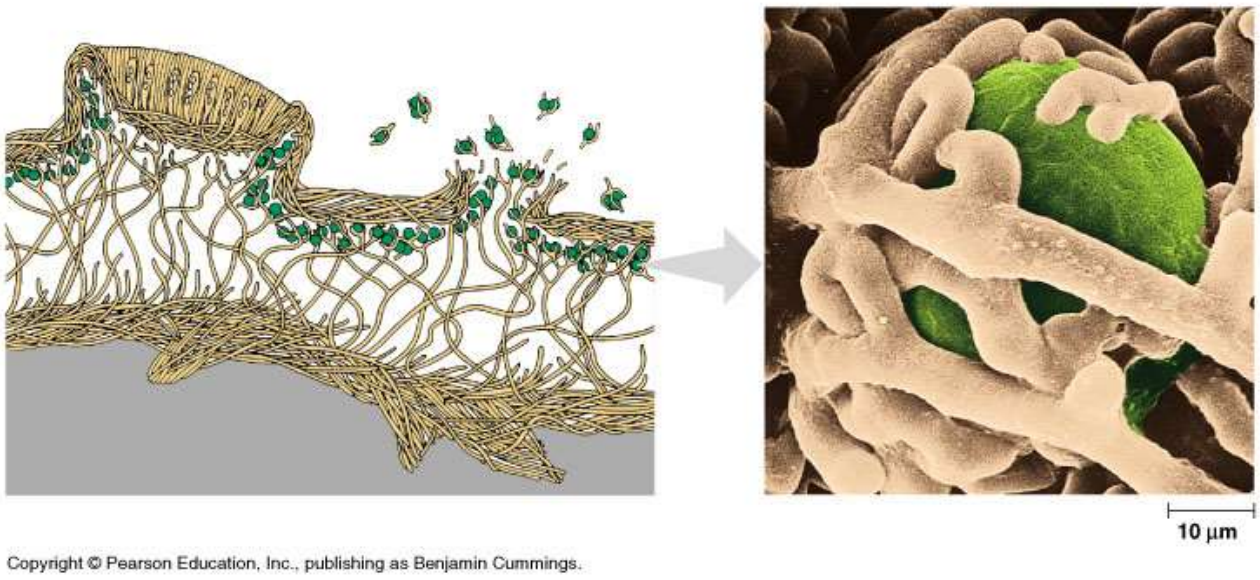


Fungal hyphae arou nd root and between cells

Lichens

- "Mutualism" between
  - Fungus – structure
  - Alga or cyanobacterium – provides food
- Three main types of lichens:
  - **Crustose lichens** form flat crusty plates.
  - **Foliose lichens** are leafy in appearance, although lobed or branched structures are not true leaves.
  - **Fruticose lichens** are even more finely branched and may hang down like beards from branches or grow up from the ground like tiny shrubs.

Lichen internal structure



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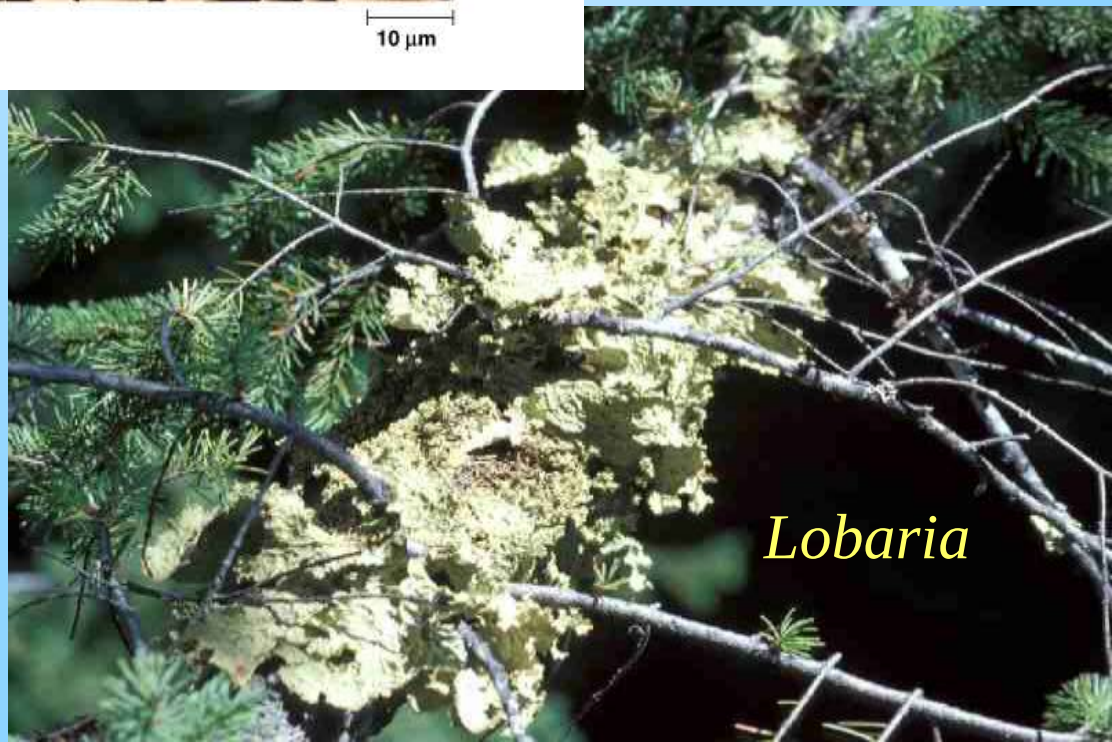
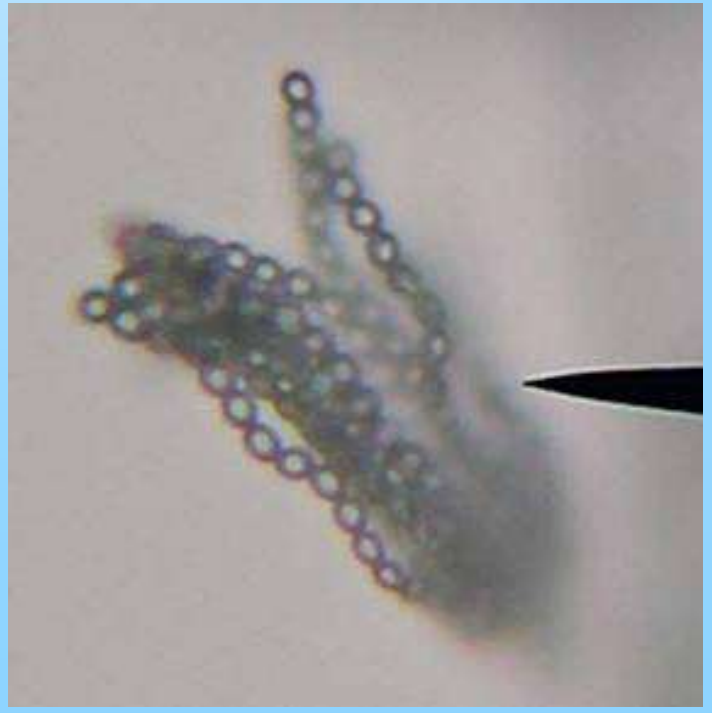
Lichens are nature's biological monitors o f pollution and air quality

- **Thalli act like sponges**
- **Some species more sensitive to pollu tion**
- **Which species are present can indic ate air quality**
- **Most resistant species can also be an alyzed for pollutants, including bioac cumulation of heavy metals and radi oactive isotopes**



Fungi are Spore-ific!!!

- Spores - asexual (product of mitosis) or sexual (product of meiosis) in origin.
- Purpose of Spores
  * Allows the fungus to move to new food source.
  * Resistant stage - allows fungus to survive periods of adversity.
  * Means of introducing new genetic combinations into a population



Fungi as Parasites & Pathogens

Reproduce by spores

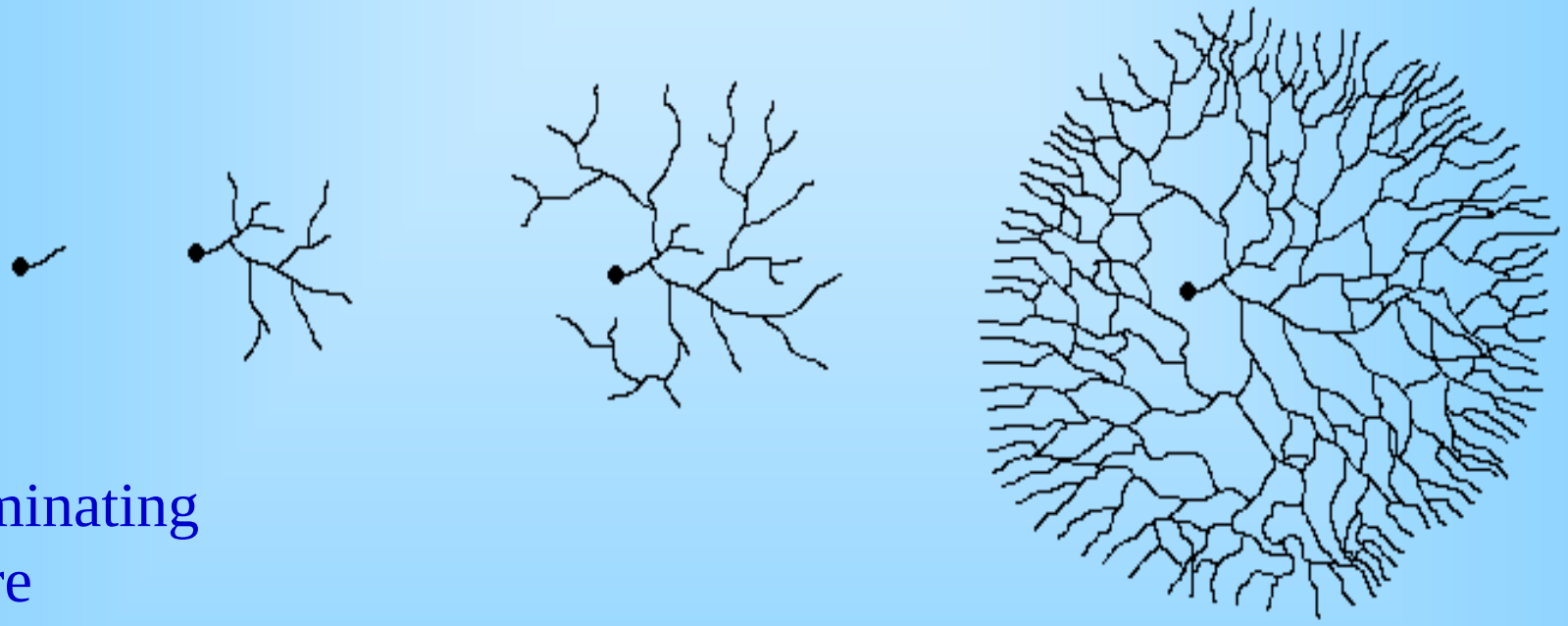
- Spores are reproductive cells
  - Sexual (meiotic in origin)
  - Asexual (mitotic in origin)
- Formed:
  - Directly on hyphae
  - Inside sporangia
  - Fruiting bodies

Amanita fruiting body

Pilobolus sporangia

Penicillium hyphae with conidia

Hyphal growth from spore



germinating spore

mycelium

- Mycelia have a huge surface area

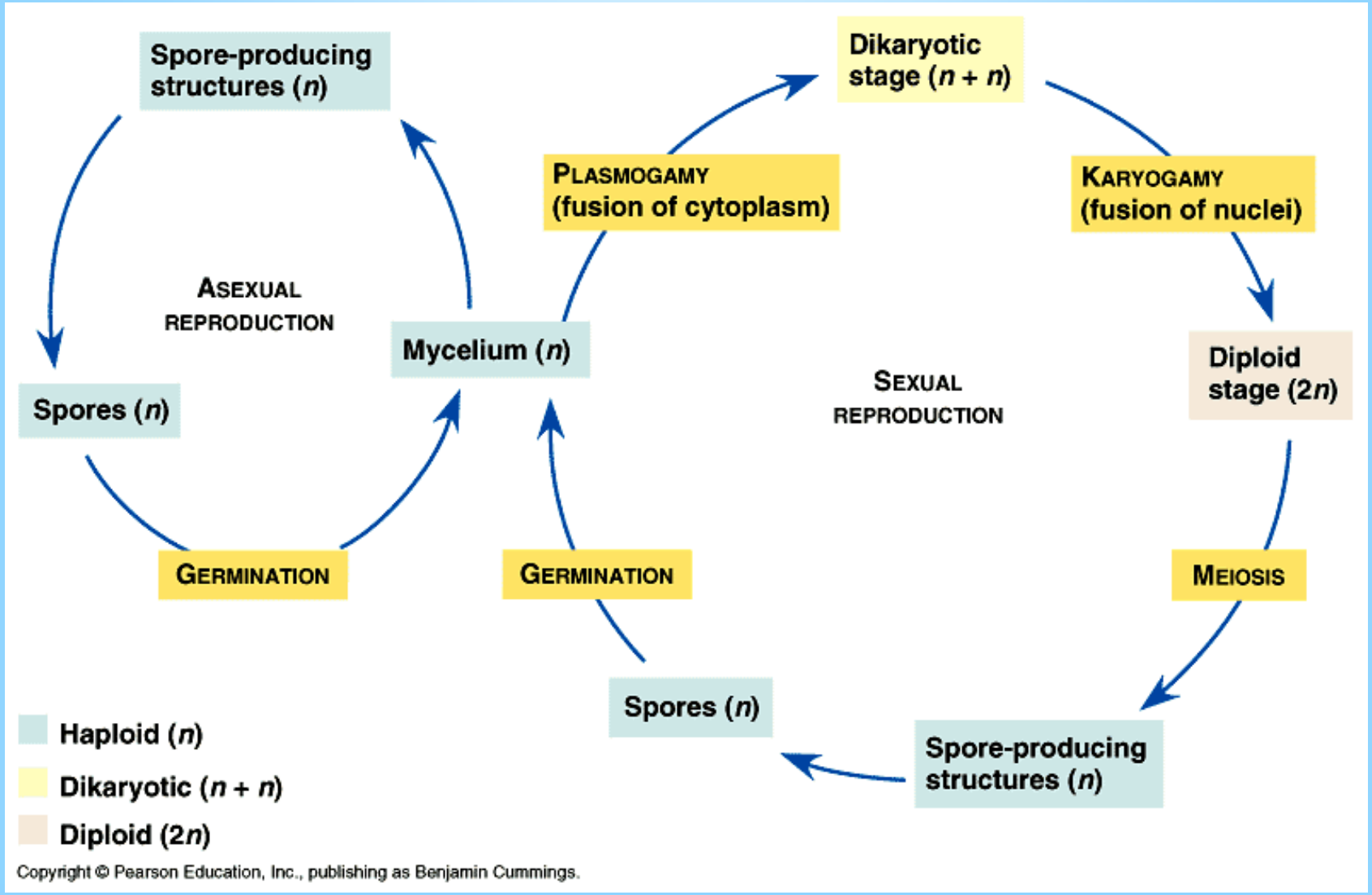
The Characteristics of Fungi

- Fungus is often hidden from view. It grows through its food source (substratum), excret es extracellular digestive enzymes, and abso rbs dissolved food.
- Indeterminate clonal growth.
- Vegetative phase of fungus is generally sed entary.

The Characteristics of Fungi

- Cell wall present, composed of cellulose and/or chitin.
- Food storage - generally in the form of lipids and glycogen.
- Eukaryotes - true nucleus and other organelles present.
- All fungi require water and oxygen (no obligate anaerobes).
- Fungi grow in almost every habitat imaginable, as long as there is some type of organic matter present and the environment is n ot too extreme.
- Diverse group, number of described species is somewhere betw een 69,000 to 100,000 (estimated 1.5 million species total).

Generalized Life Cycle of a Fungus



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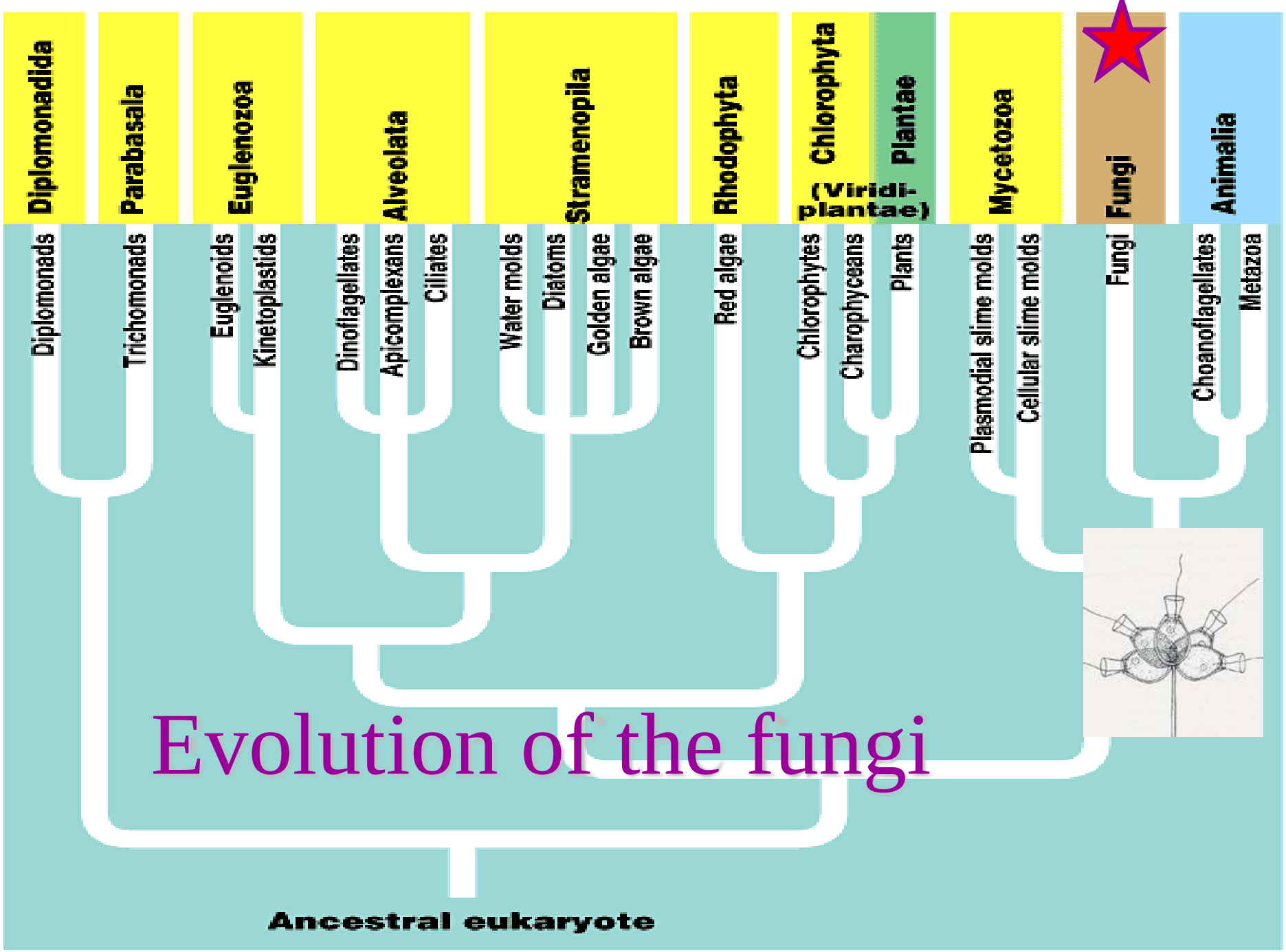
Evolution of the fungi



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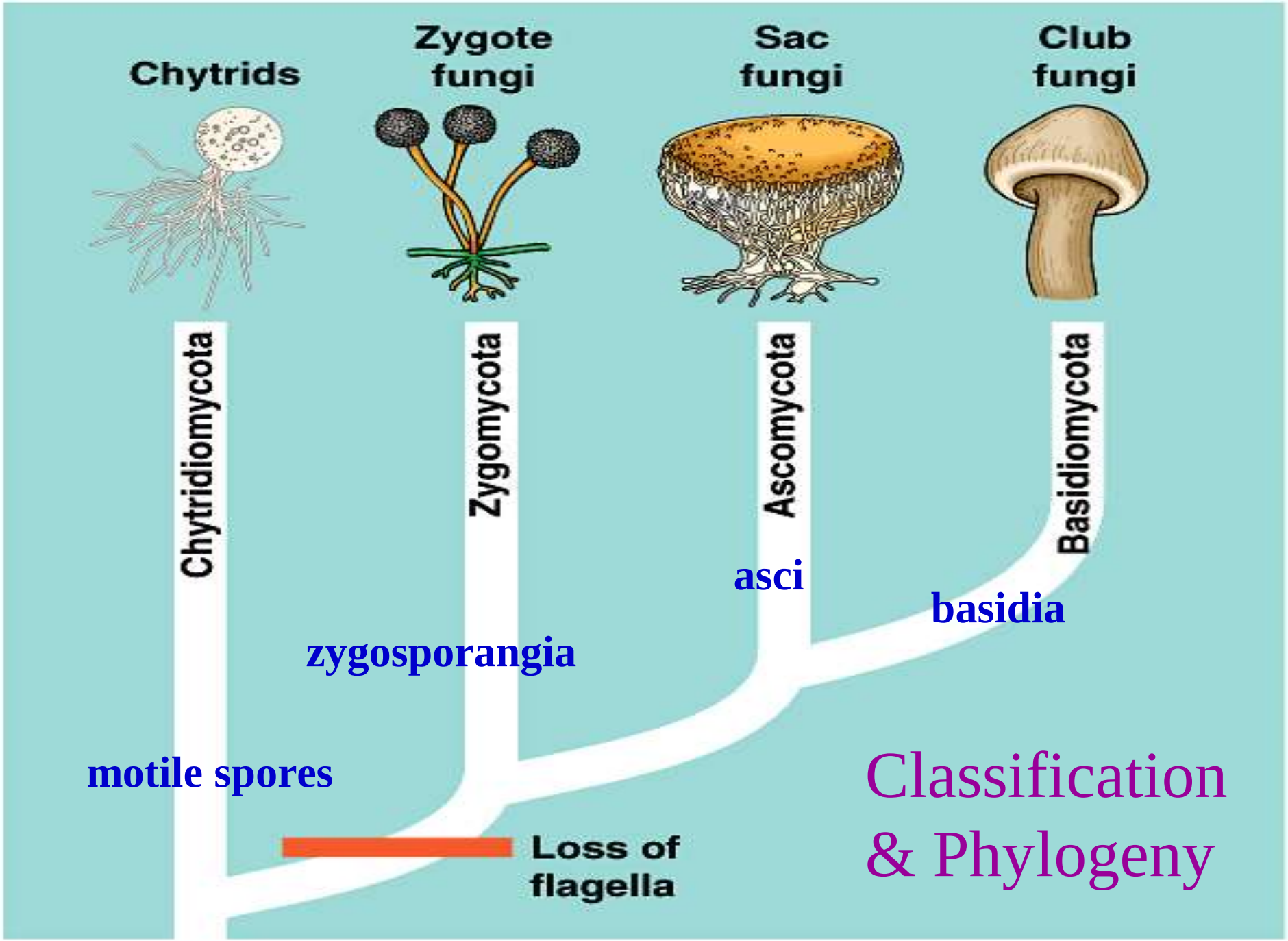

Classification & Phylogeny

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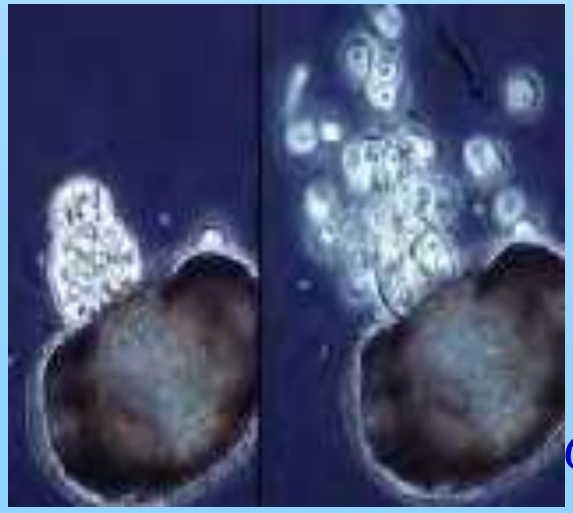
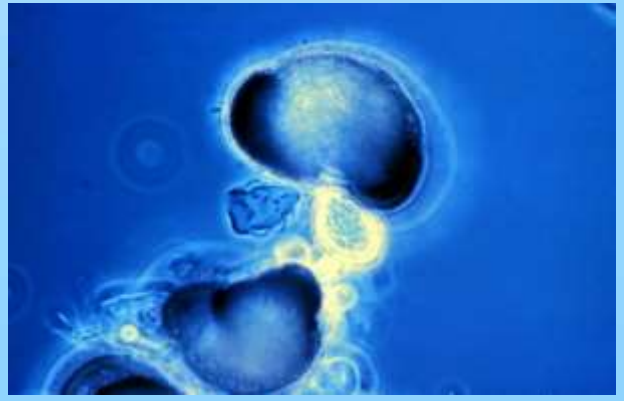
Chytridiomycota – “chytrids”

- Simple fungi
- Produce motile spores - zoospores
- Mostly saprobes and parasites in aquatic habitats
- Could just as well be Protists

Chytridium growing on spores





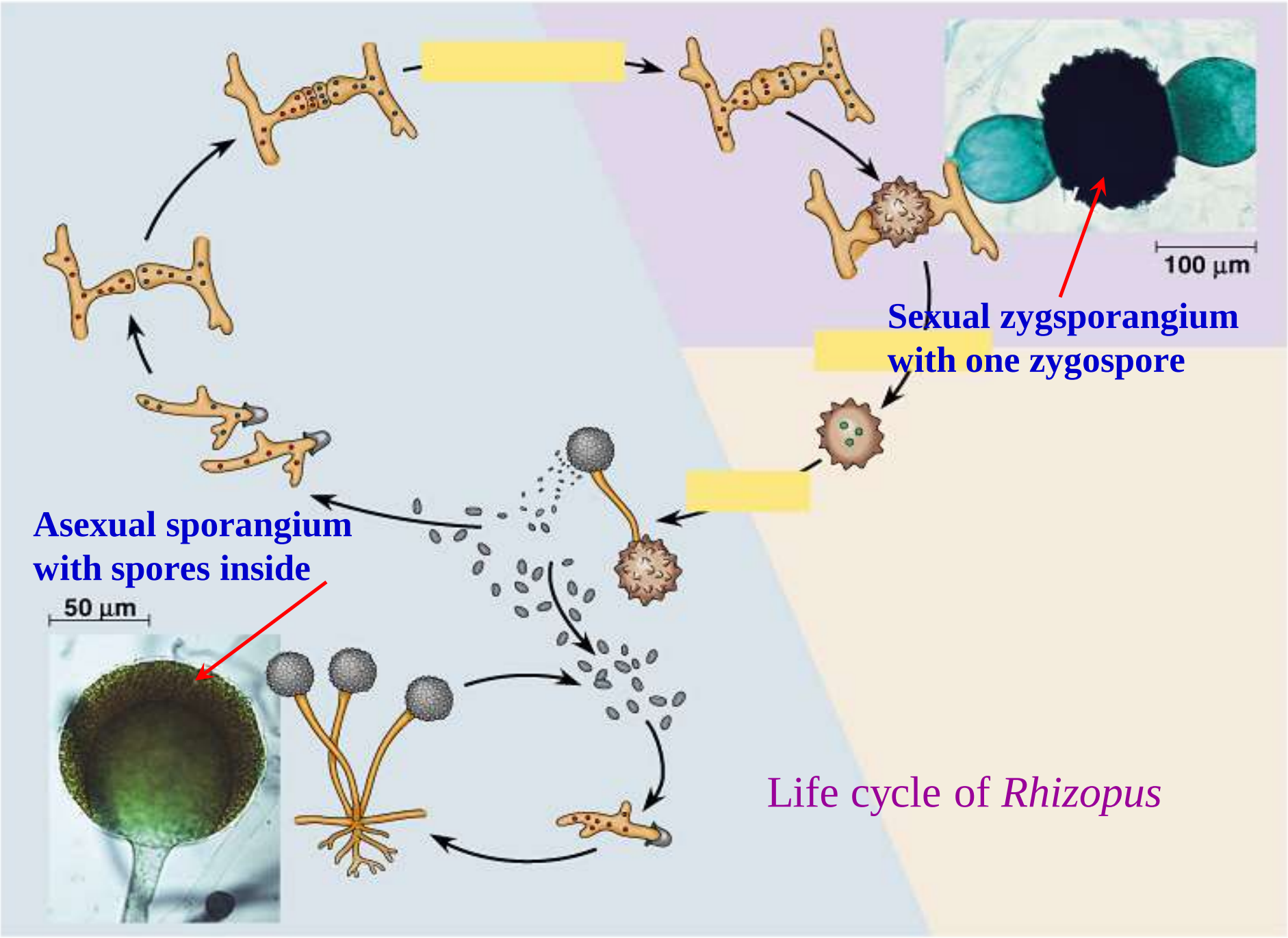
Chytriomyces growing on pine pollen

Zygomycota – “zygote fungi”

Rhizopus on strawberries

- Sexual Reproduction - zygosporangia
- Asexual reprod. – common (sporangia – bags of asexual spores)
- Hyphae have no cross walls
- Grow rapidly
- Decomposers, pathogens, and some form mycorrhizal associations with plants

Rhinocerebral zygomycosis



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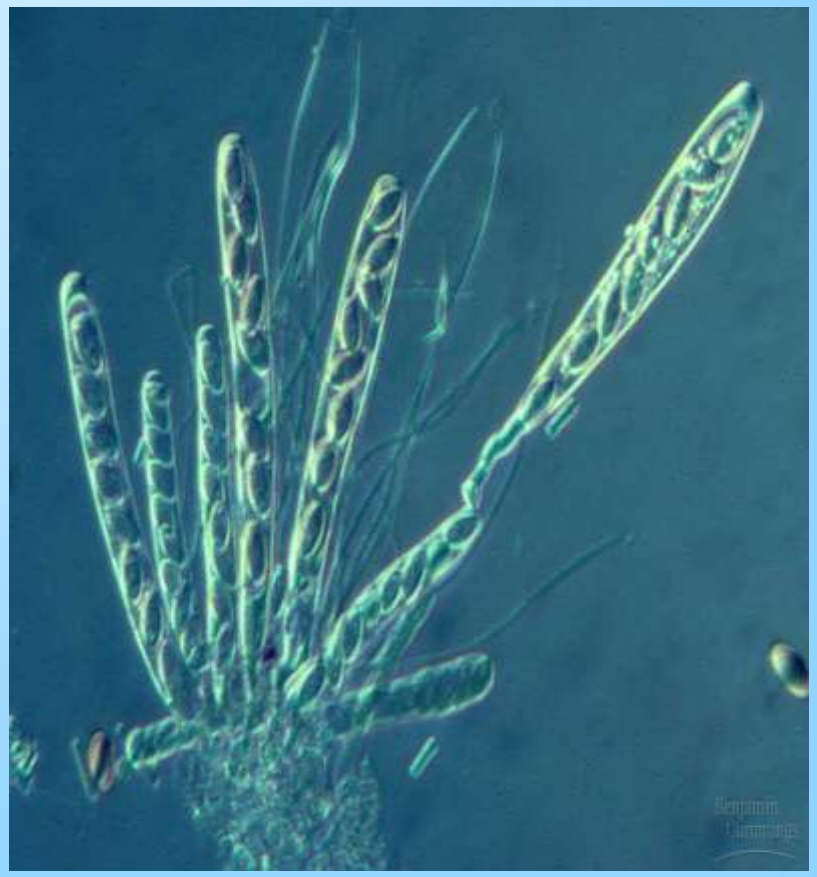
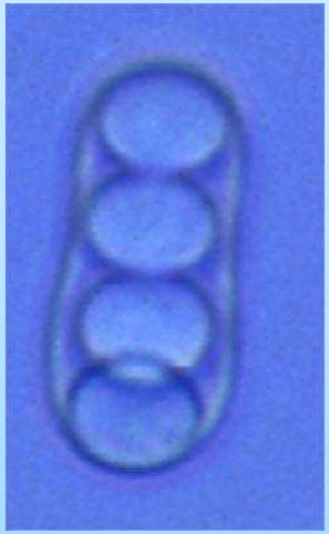
Ascomycota – “sac fungi”

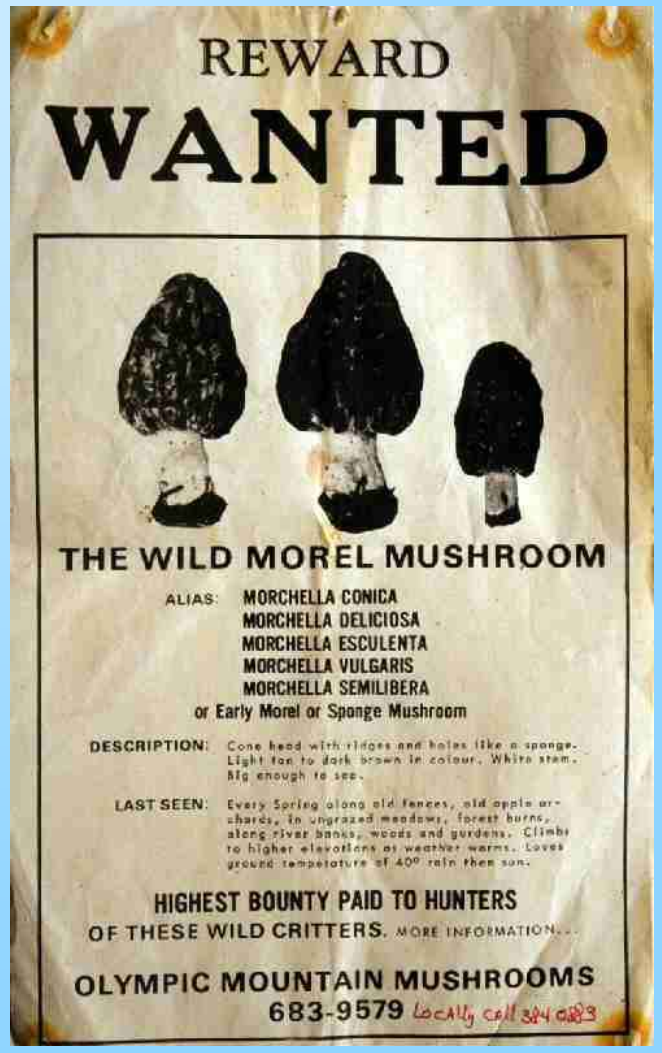
- Sexual Reproduction – asci (sing. = ascus)
- Asex. Reprod. – common
- Cup fungi, morels, truffles
- Important plant parasites & saprobes
- Yeast - *Saccharomyces*
- Decomposers, pathogens, and found in most lichens



A cluster of asci with spores inside

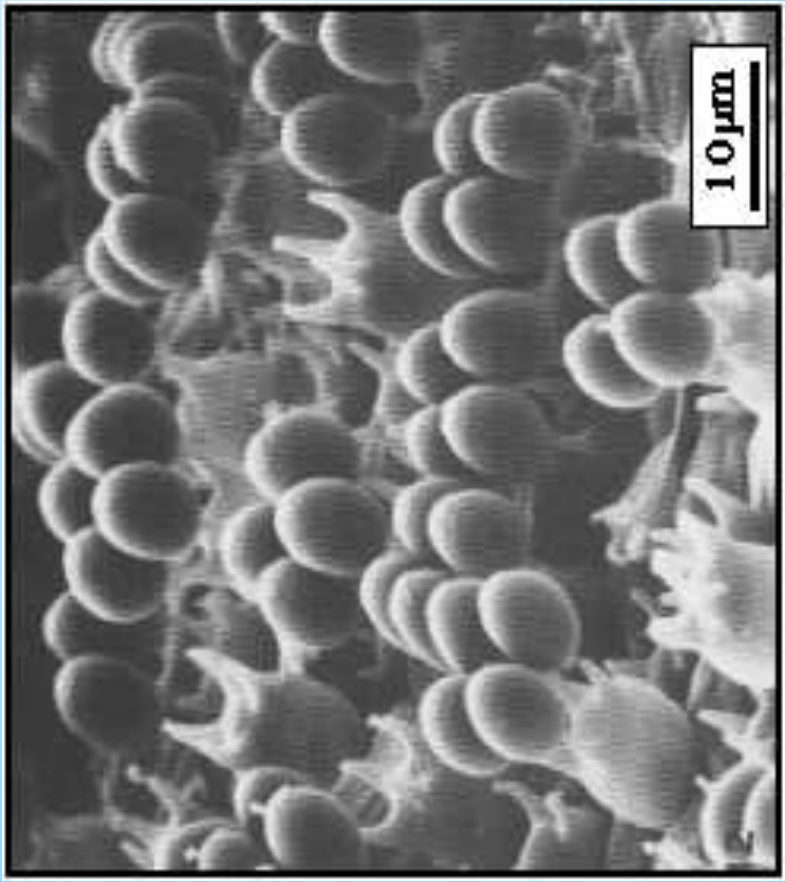
Sac fungi diversity





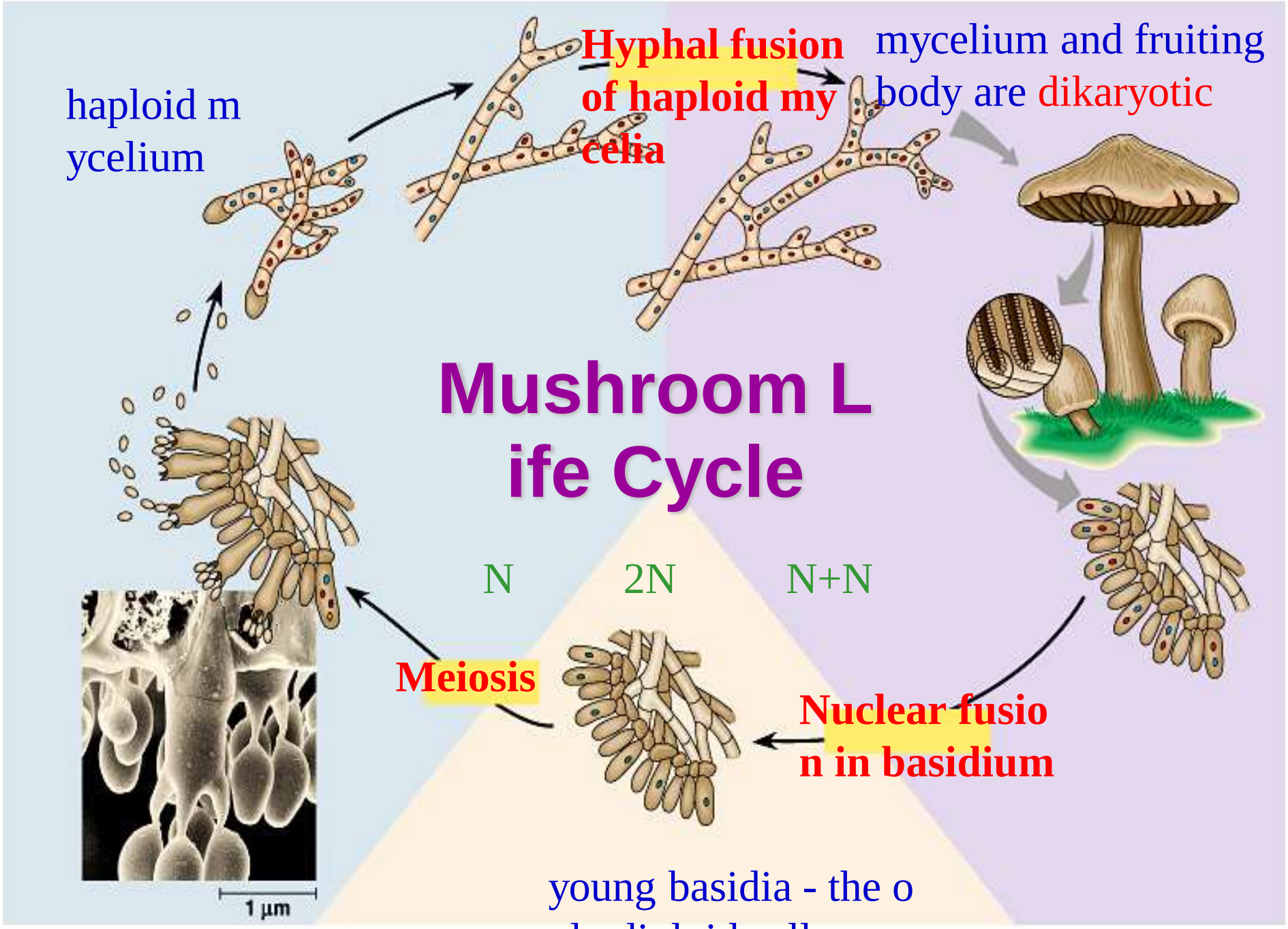
Basidiomycota – “club fungi”

- Sexual Reproduction – basidia
- Asexual reprod – not so common
- Long-lived **dikaryotic** mycelia
- Rusts & smuts –plant parasites
- Mushrooms, polypores, puffballs, boletes, bird’s nest fungi
- Enzymes decompose wood, leaves, and other organic materials
- Decomposers, pathogens, and some form mycorrhizal associations with plants



SEM of basidia and spores

Mushroom L ife Cycle



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Bioluminescence in *Mycena*

Some fungi have more than on e scientific name – Why?

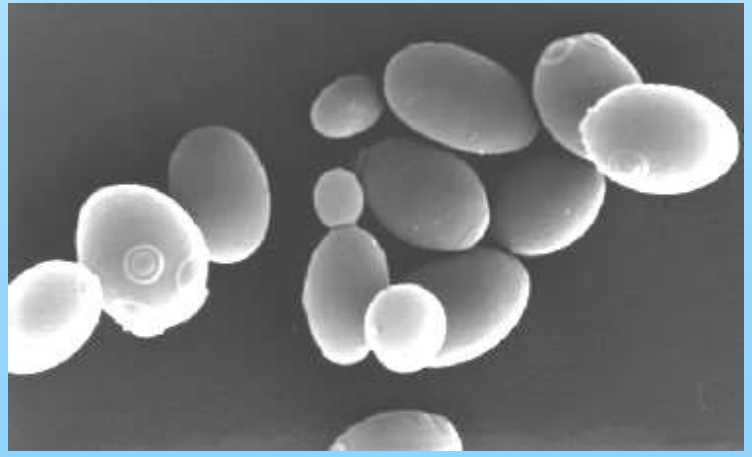
- **Teleomorph**: the sexual reproductive stage (morp h), typically a fruiting body (e.g., *Morchella escul enta*, *Agaricus brunescens*).
- **Anamorph**: an asexual reproductive stage (morph), often mold-like (e.g. *Aspergillus flavus*, *Fusariu m solani*). When a single fungus produces multipl e morphologically distinct anamorphs, they are cal led **synanamorphs**.
- **Holomorph**: the whole fungus, including all anam orphs and the teleomorph.

Deuteromycota – Form Phylum "Imperfect Fungi"

- Fungi that seldom or never reproduce sexua lly.
- Asexual reproduction by vegetative growth and production of asexual spores common.

Yeasts

- Single celled fungi
- Adapted to liquids
  - * Plant saps
  - * Water films
  - * Moist animal tissues



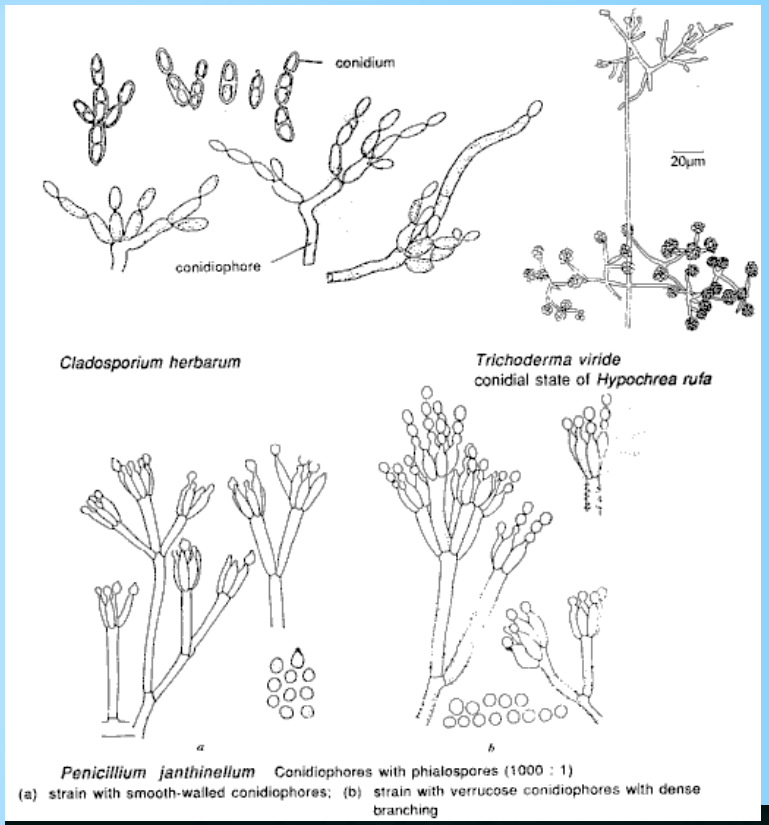
Saccharomyces

Candida

Molds

- Rapidly growth
- Asexual spores
- Many human importances
  - * Food spoilage
  - * Food products
  - * Antibiotics, etc.

Noble Rot - *Botrytis*



HUMAN-FUNGUS INTERACTIONS

- **Beneficial Effects of Fungi**
  - Decomposition - nutrient and carbon recycling.
  - Biosynthetic factories. Can be used to produce drugs, antibiotics, al cohol, acids, food (e.g., fermented products, mushrooms).
  - Model organisms for biochemical and genetic studies.
- **Harmful Effects of Fungi**
  - Destruction of food, lumber, paper, and cloth.
  - Animal and human diseases, including allergies.
  - Toxins produced by poisonous mushrooms and within food (e.g., g rain, cheese, etc.).
  - Plant diseases.